
CAPYTAINE
([HTTPS://CAPYTAINE.GITHUB.IO/STABLE](https://capytaine.github.io/stable))

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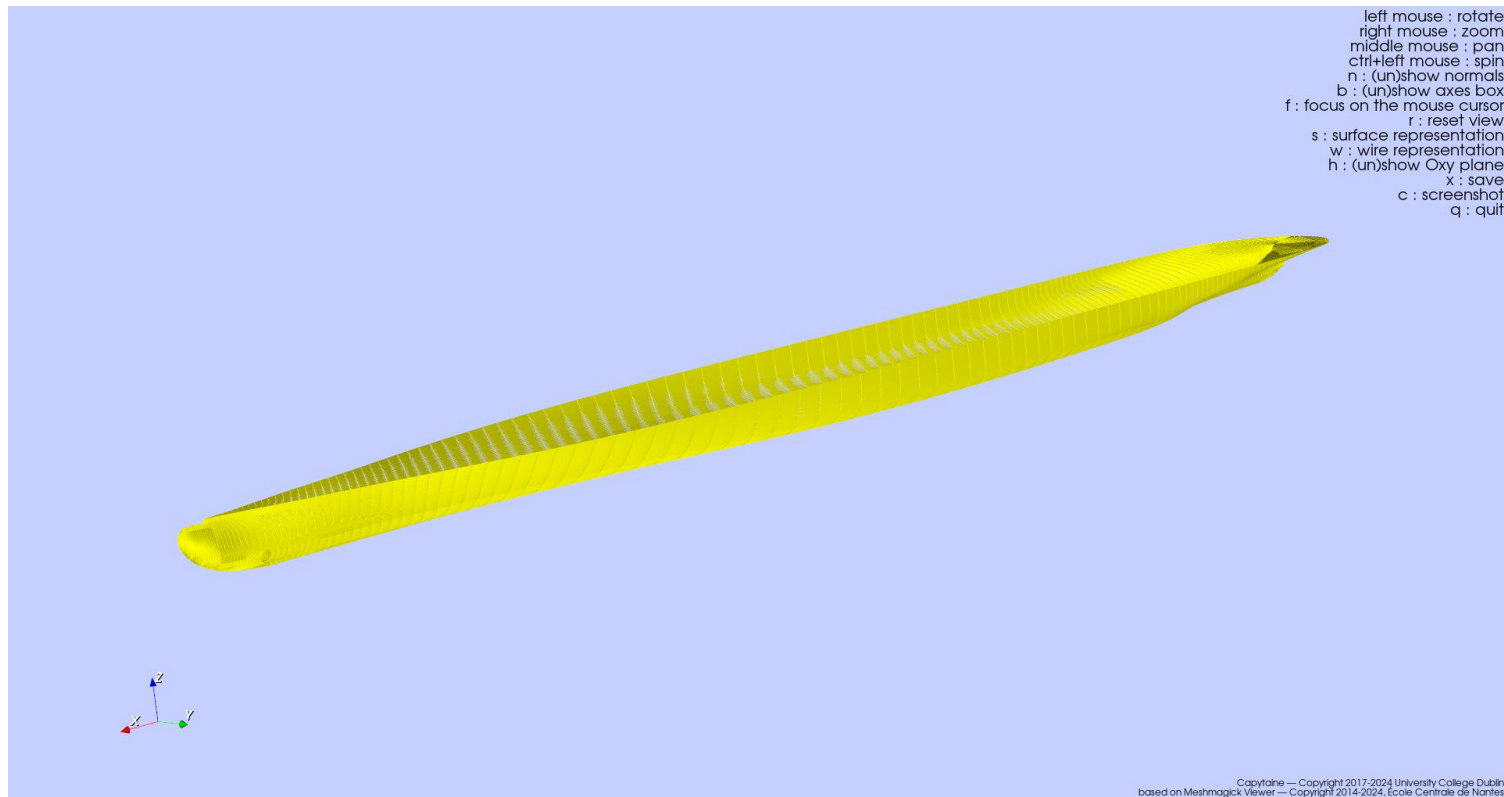
FOREWORD

Capytaine is a Python package for the simulation of the interaction between water waves and floating bodies in the frequency domain. It is built around a full rewrite of the open source Boundary Element Method (BEM) solver Nemoh for the linear potential flow wave theory.

Version 2.2 of **Capytaine** (<https://capytaine.github.io/stable/>) was launched recently and features the approximate forward speed (AFS) based on the work described by Donatini et al. (2022), which was published in the proceedings of the 6th MASHCON conference. **Capytaine**, developed by Matthieu Ancellin (R&D Engineer at Eurobios Mews Labs <https://www.mews-partners.com/en/our-offering/eurobios-mews-labs>), is an open-source linear potential flow solver capable of simulating wave loads on structures in the frequency domain. Examples of applications include seakeeping calculations of floating objects or structures. In the AFS approach, the zero speed results (i.e. diffraction and radiation forces) are corrected to account for the fluid velocities (parallel to the direction of the forward speed) considering the diffracted and radiated waves. The AFS approach assumes that the magnitude of the forward speed is relatively small and assumes that thin ship theory is valid so that the fluid around the body can be approximated by uniform flow. This release is also available on <https://pypi.org/project/capytaine/>.

APPLICATION TO A 18,000 TEU CONTAINER SHIP

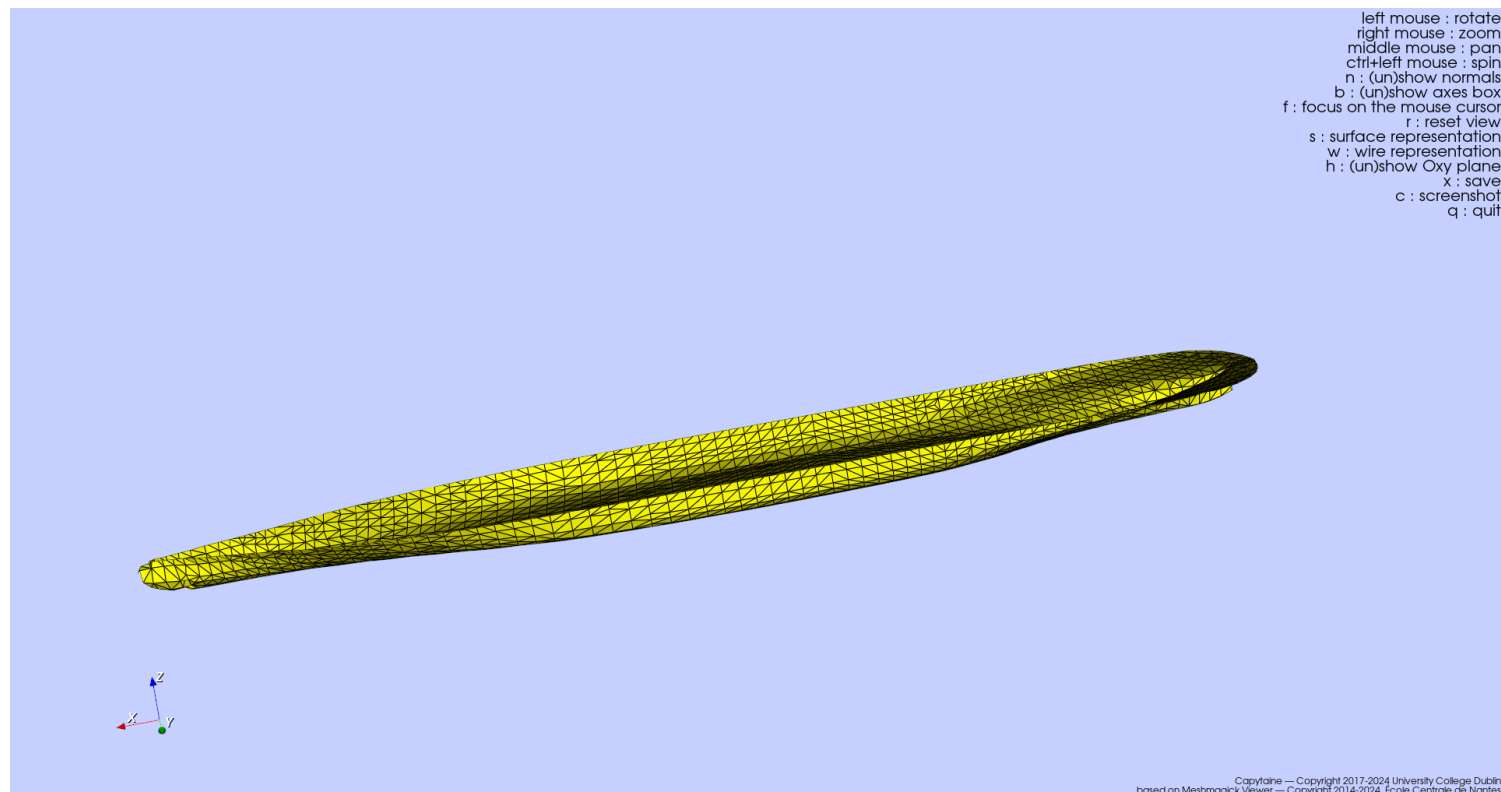
Capytaine was used to calculate the forces and movements induced by waves on a large container ship. To do this, a first STL format file of the ship was obtained with DELFTship software. With an STL format, all visible parts are exported as a large collection of small triangles. However, the file was too large to be used with a small capacity computer. It was then necessary to decimate the STL file.



Initial STL file

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One popular online tool for reducing STL file size is MeshLab. This free software allows to import the STL file and use the simplification tools to reduce the number of triangles in the mesh. By doing so, it is possible to significantly decrease the file size while maintaining the overall shape and details of the 3D model. A simplified file was then obtained by selecting the "Filters" menu and navigating to the "Remeshing, Simplification and Reconstruction" option.



Final STL file

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A module was created to obtain the different hydrostatic and hydrodynamic characteristics (masses, added masses, radiation dampings, excitation forces) as well as the RAOs.

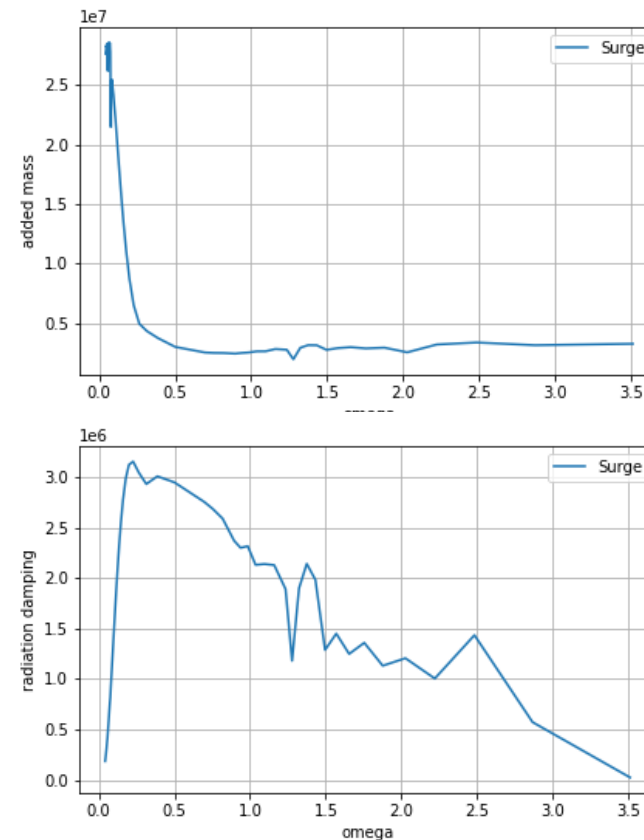
Comparison with the added masses and radiation dampings obtained with PDSTRIP is given hereafter:

Surge PDSTRIP

Added mass is constant and equal to 5.15×10^6

As PDSTRIP is a 2D strip theory program, there is no calculation of surge radiation damping. Nevertheless, in the shipmooring program (SIMBAD), an estimation of the surge radiation damping is carried out.

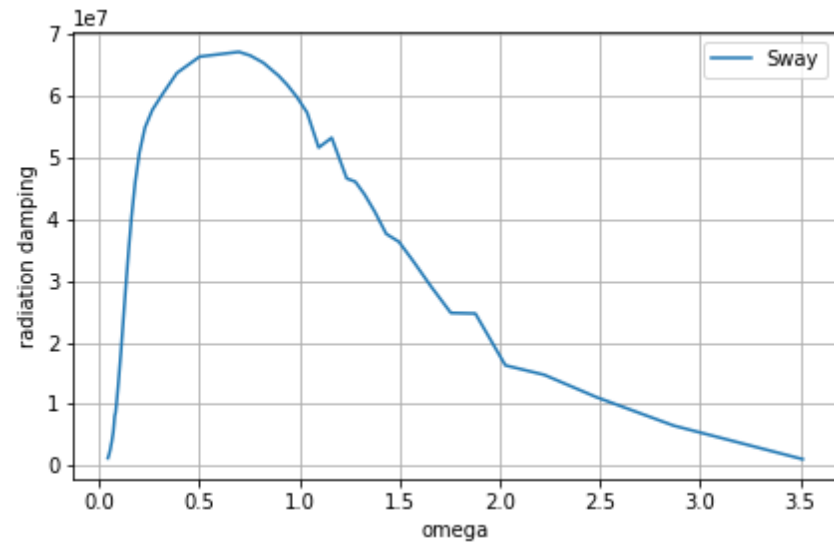
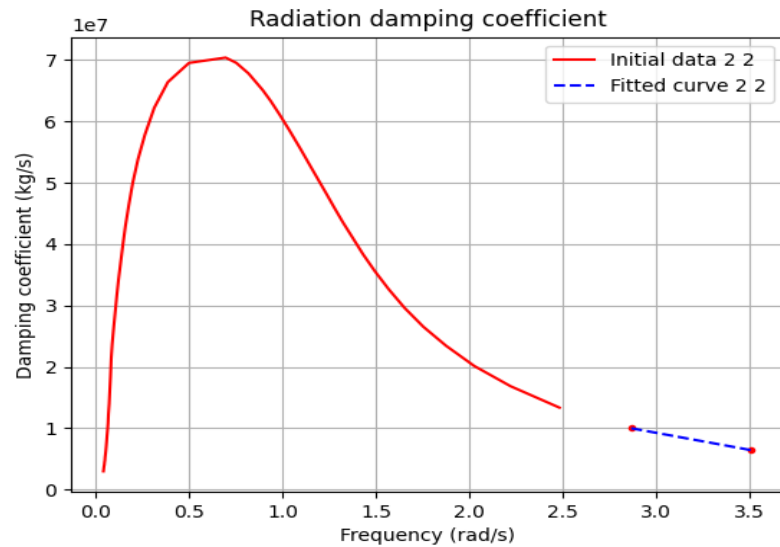
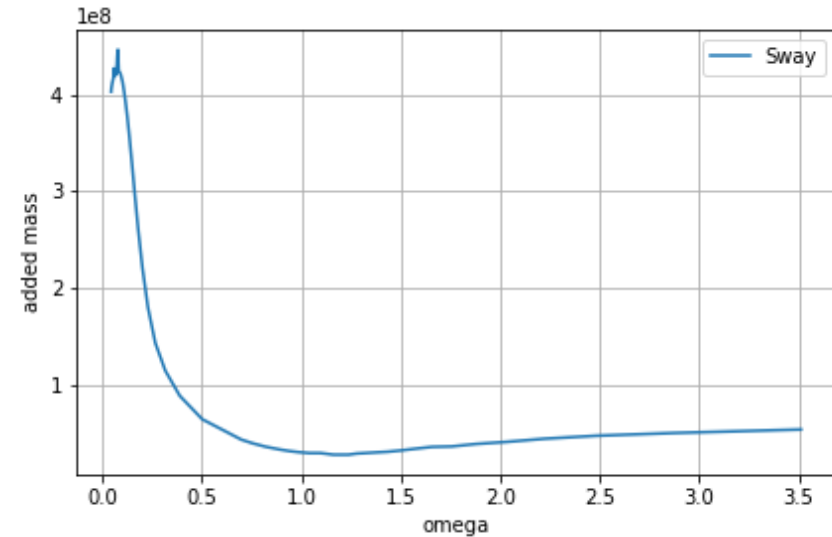
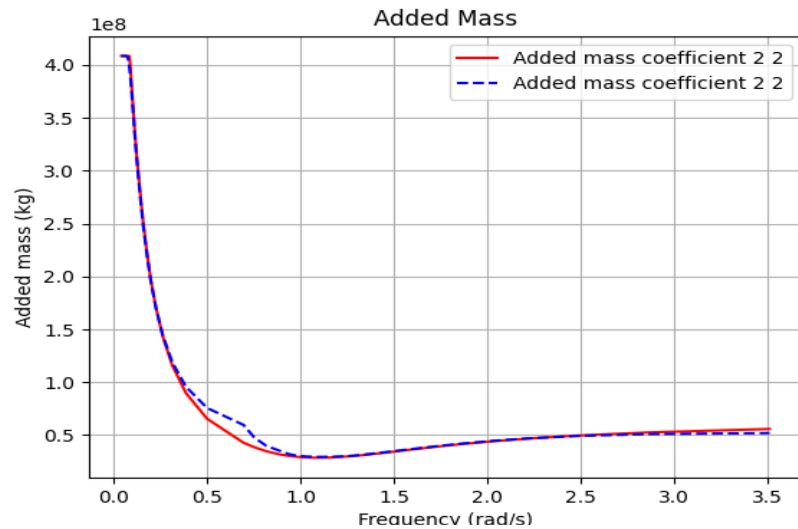
Surge CAPYTAINE



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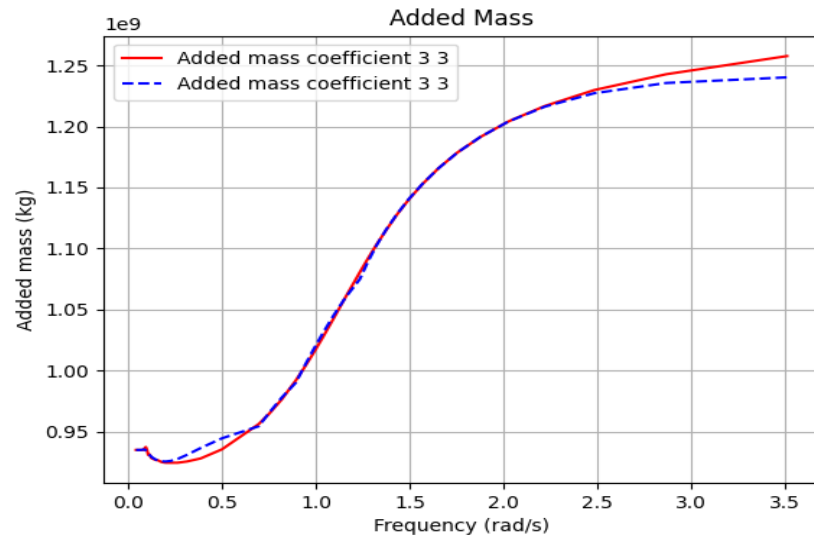
Sway PDSTRIP

Sway CAPYTAINE

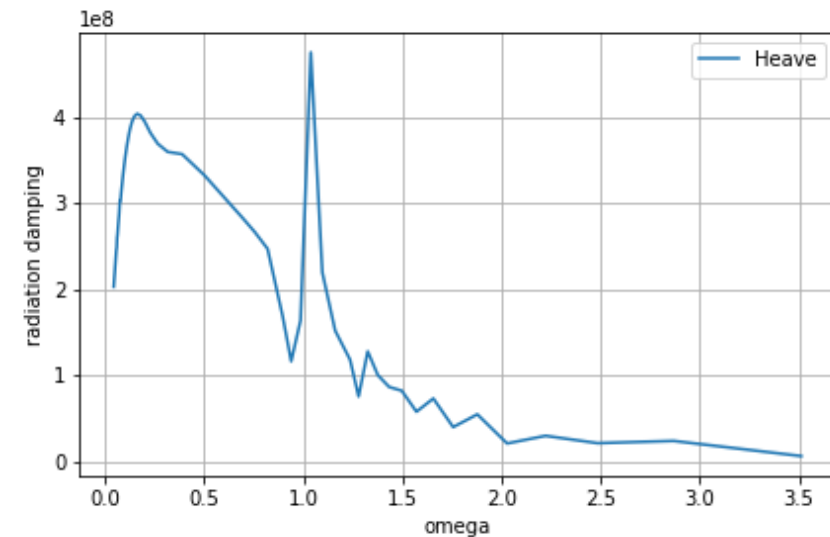
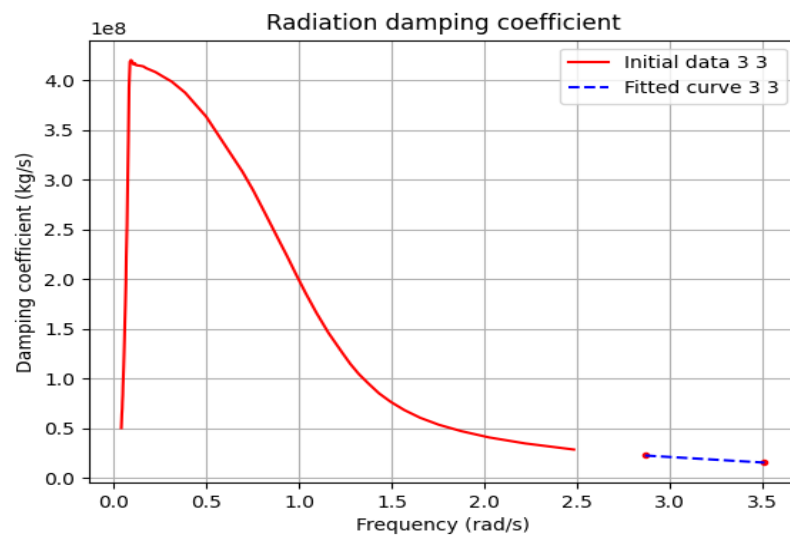
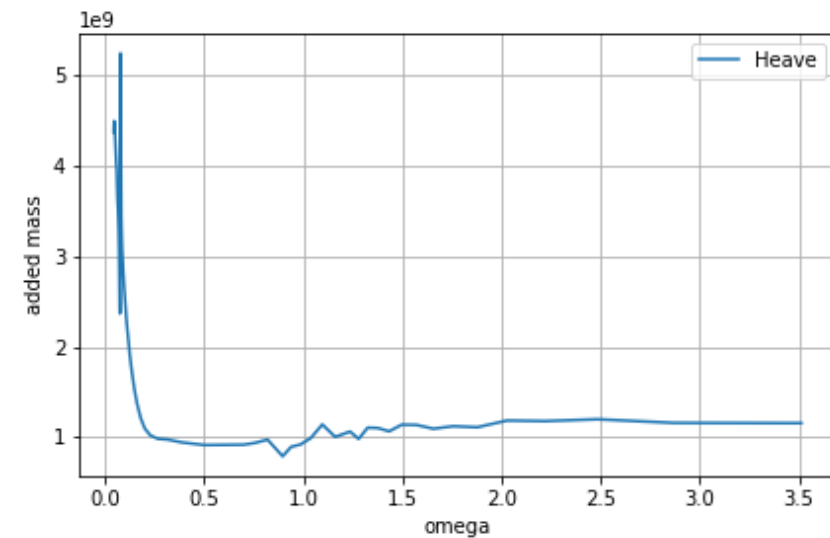


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Heave PDSTRIP

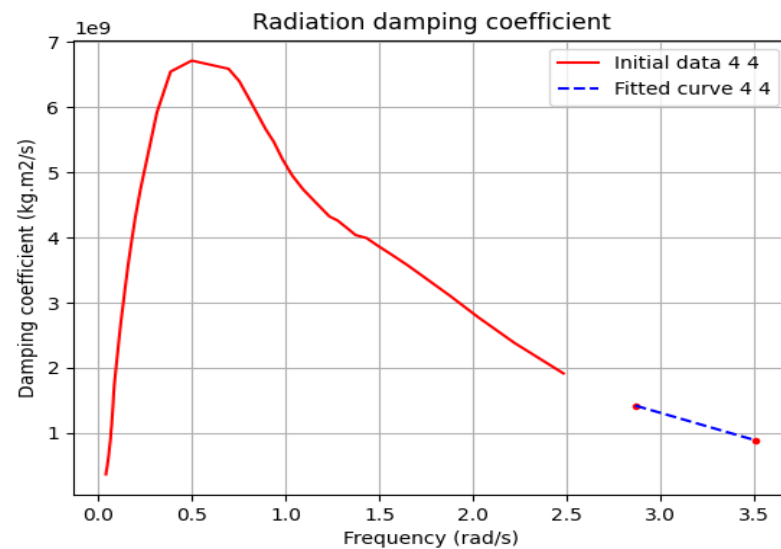
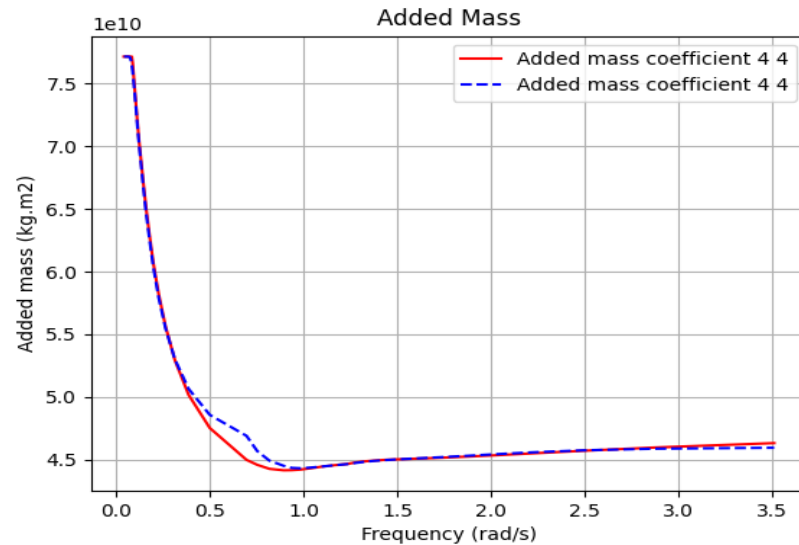


Heave CAPYTAINE

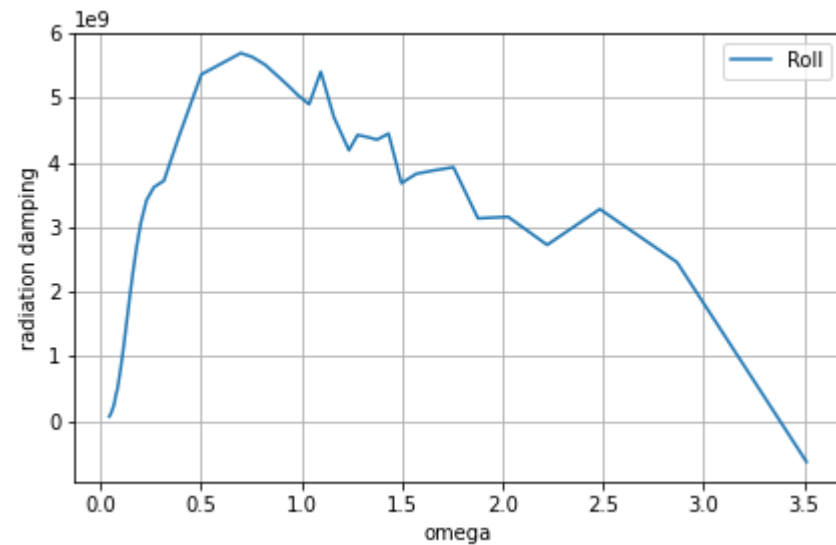
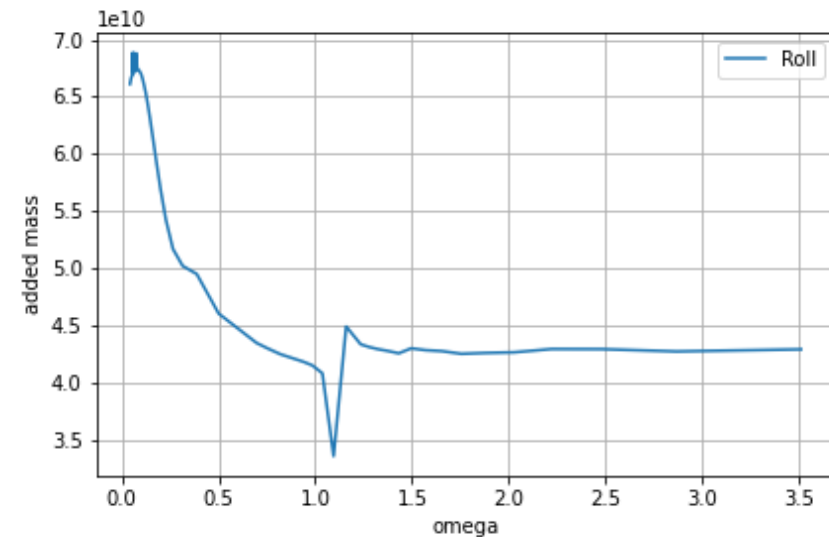


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Roll PDSTRIP

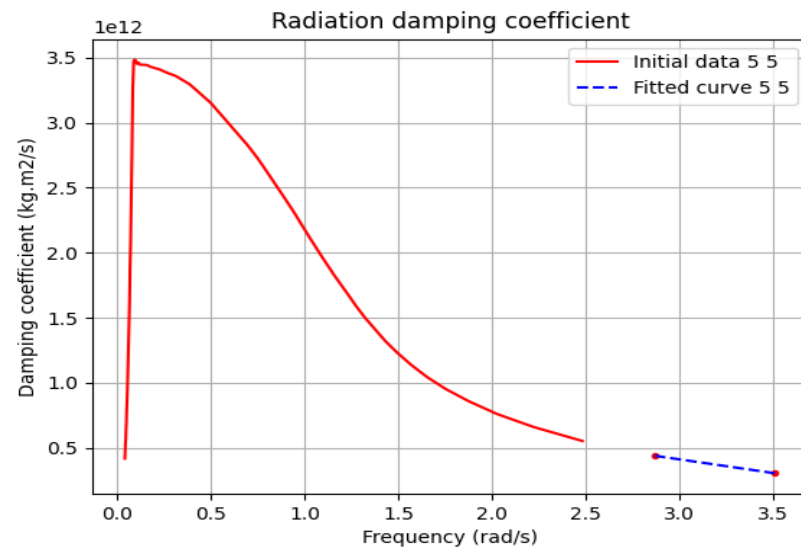
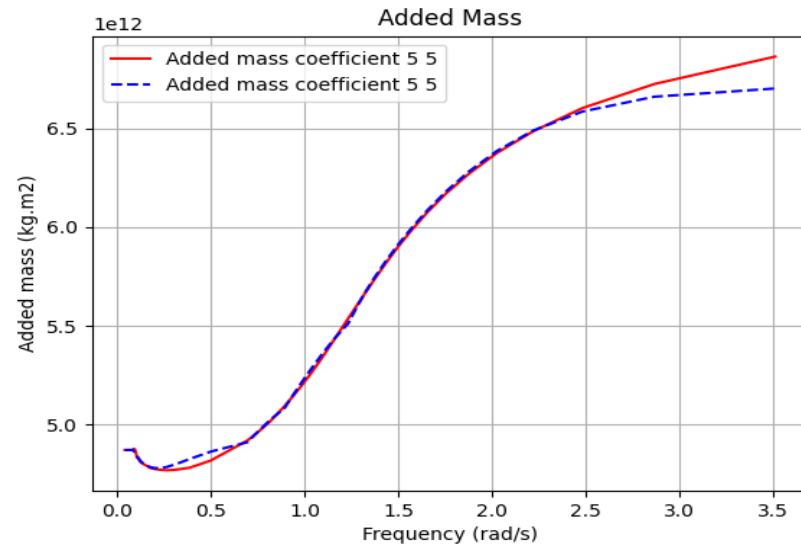


Roll CAPYTAINE

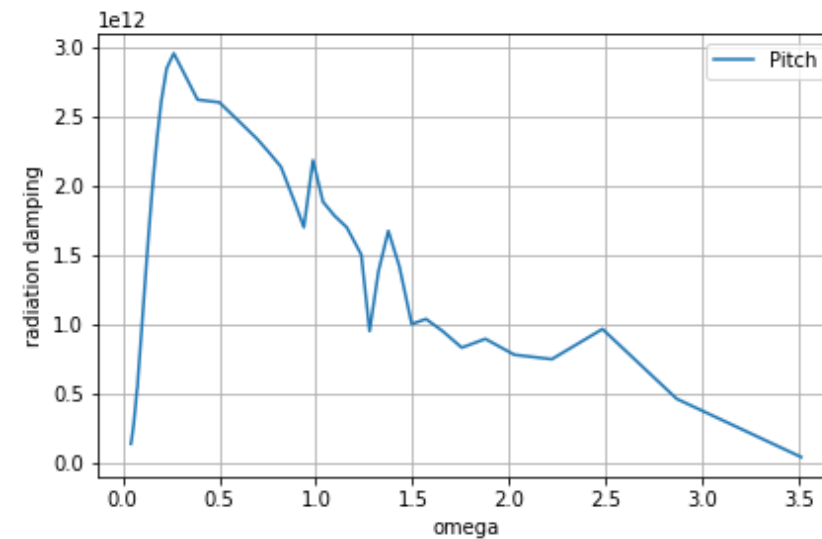
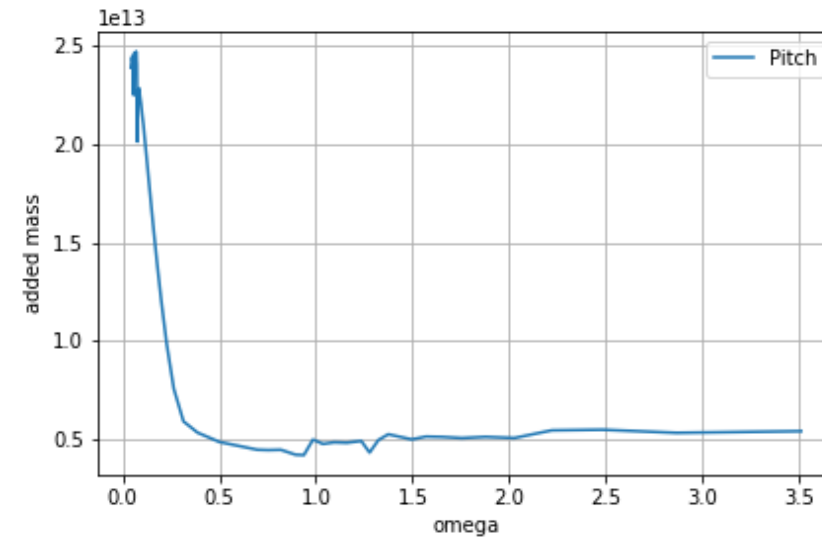


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Pitch PDSTRIP

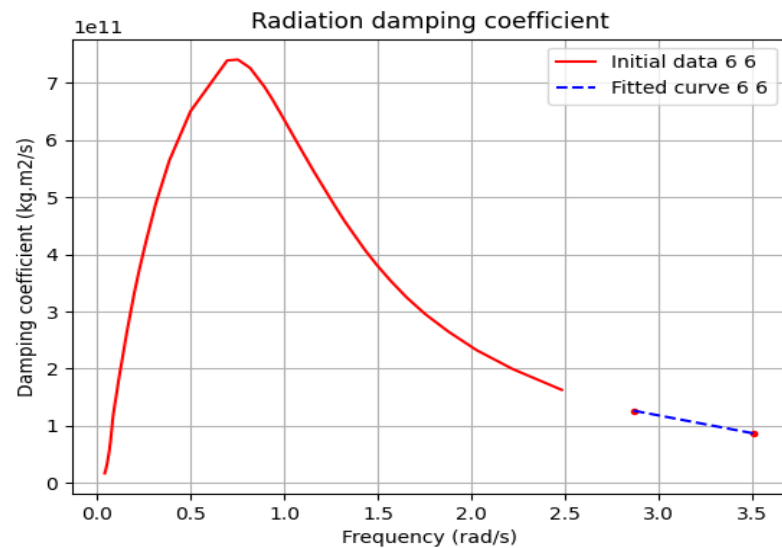
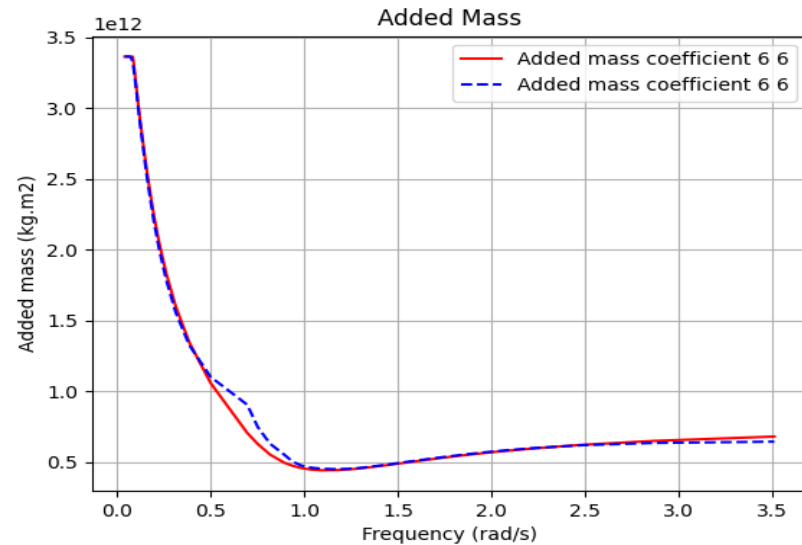


Pitch CAPYTAINE

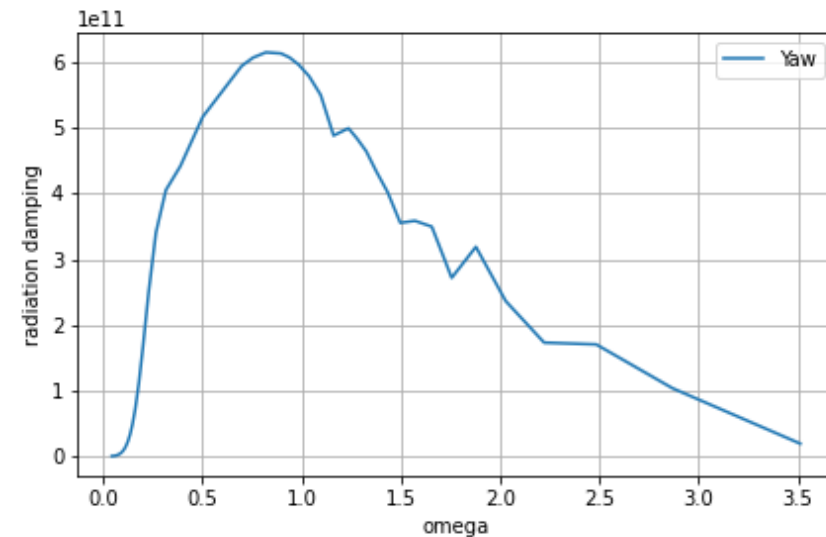
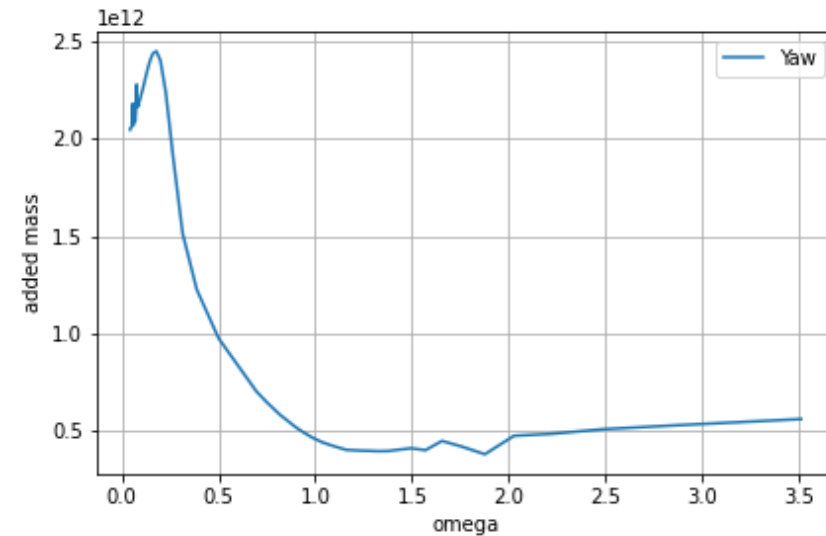


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Yaw PDSTRIP



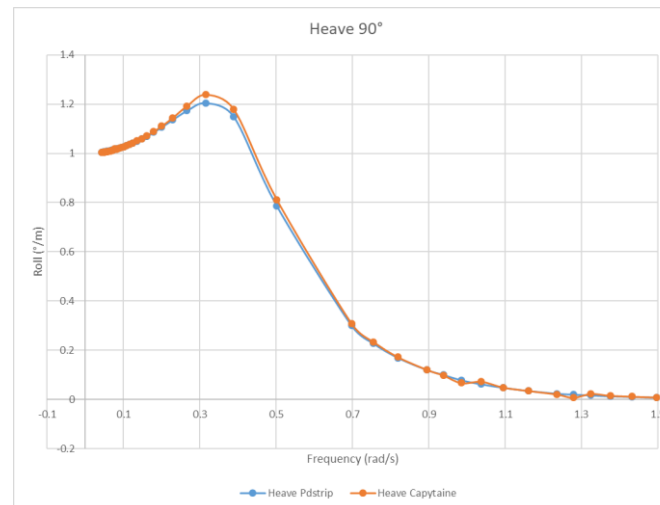
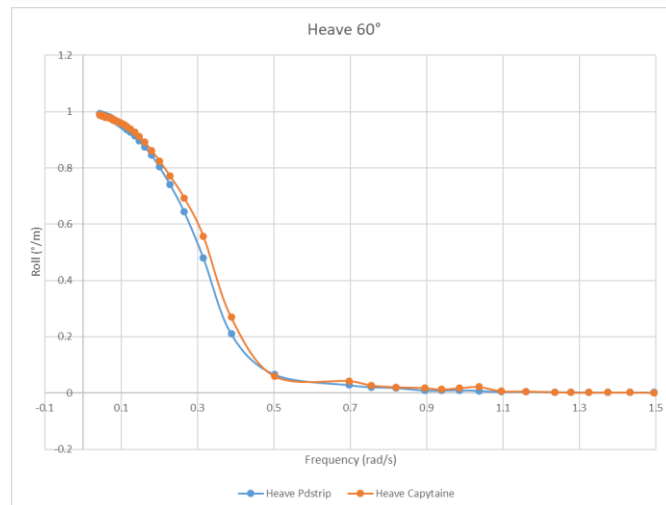
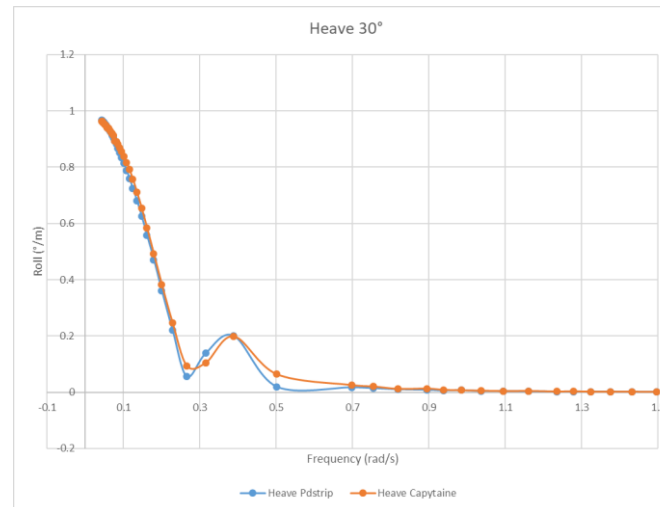
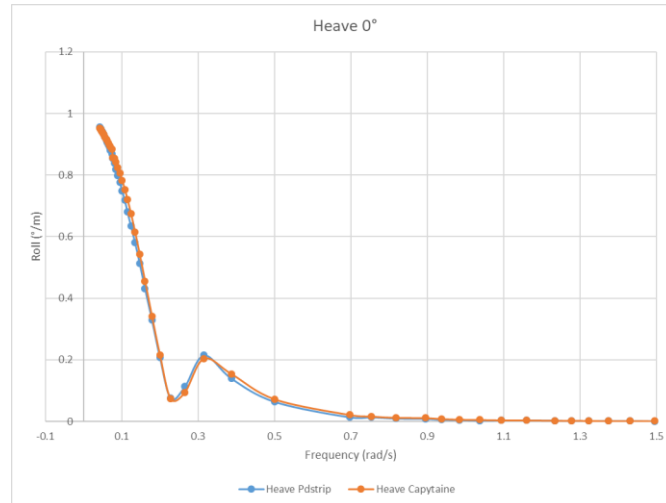
Yaw CAPYTAINE



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The comparison with the RAOs obtained with PDSTRIP shows:

- A very good agreement for the heave results

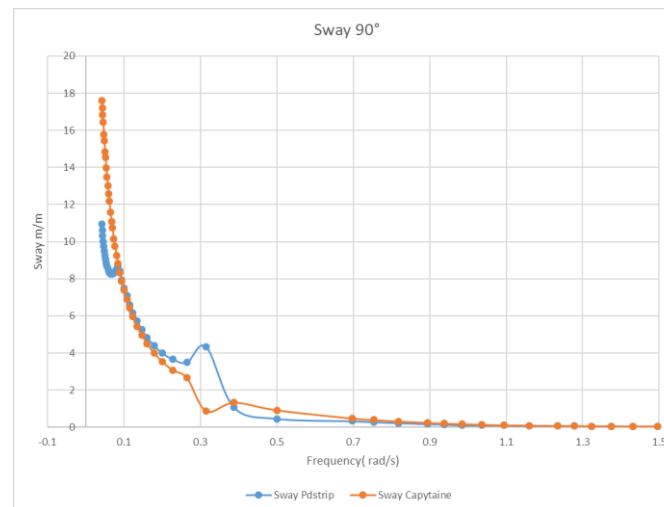
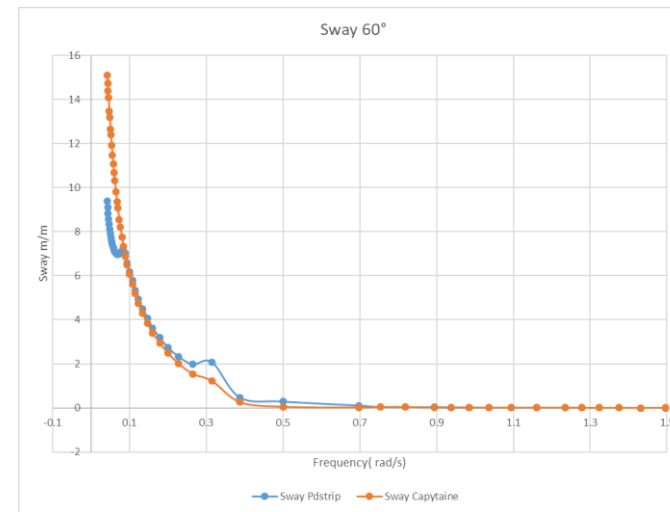
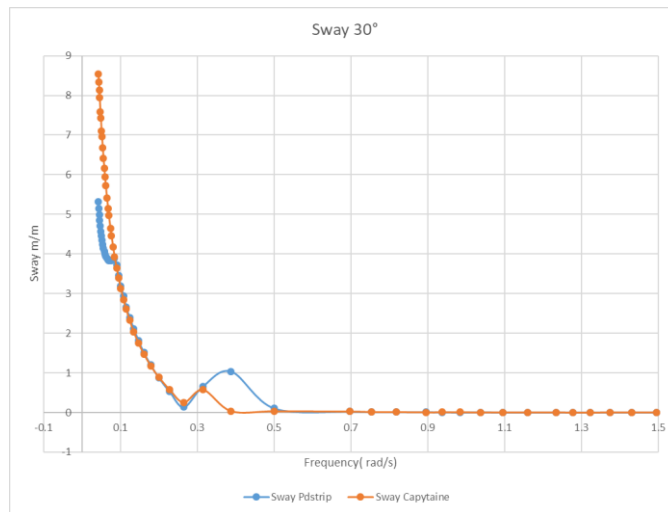


Heave results

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The comparison with the RAOs obtained with PDSTRIP shows:

- A fairly good agreement for the sway, pitch and yaw results

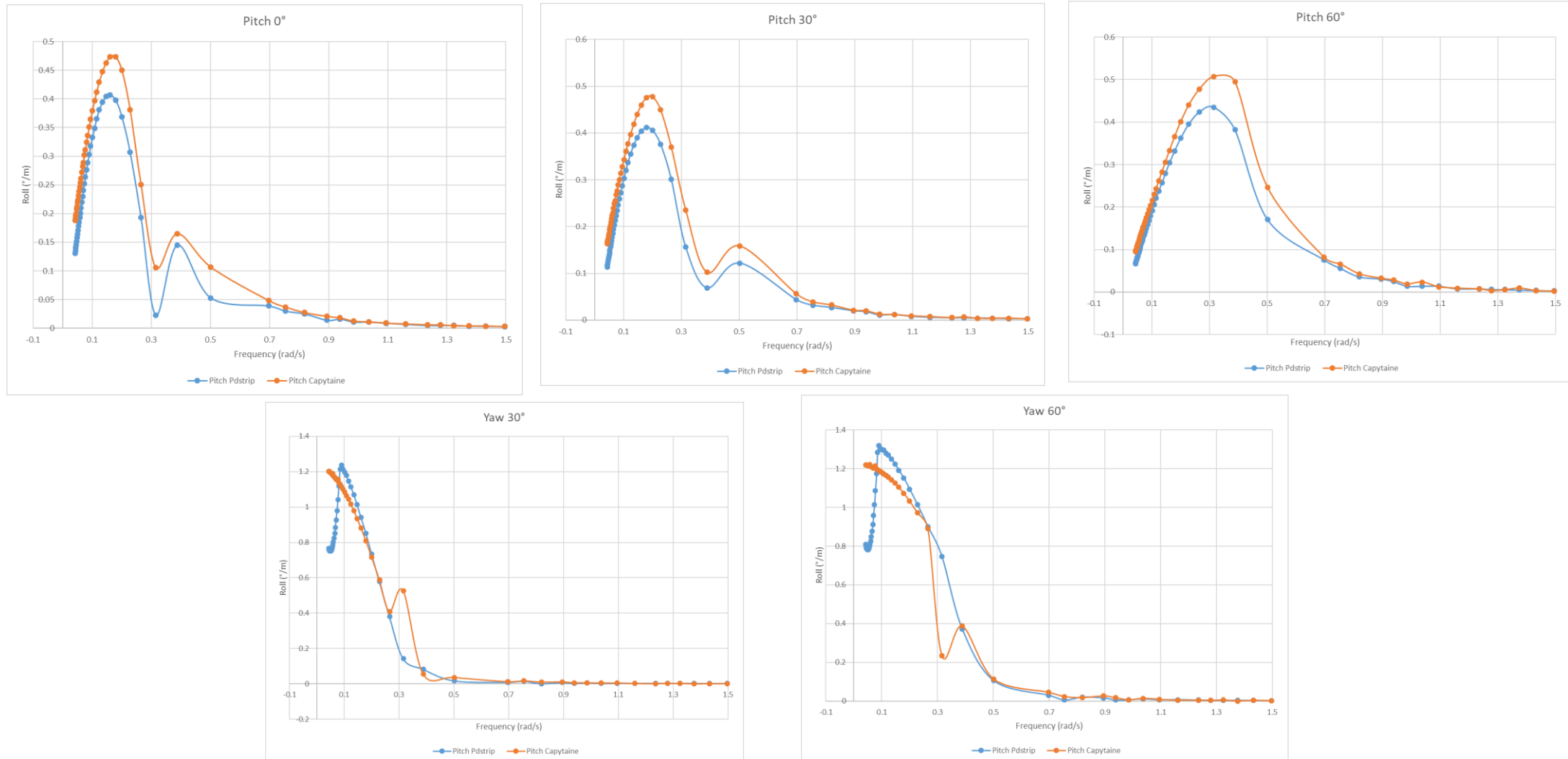


Sway results

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The comparison with the RAOs obtained with PDSTRIP shows:

- A fairly good agreement for the sway, pitch and yaw results

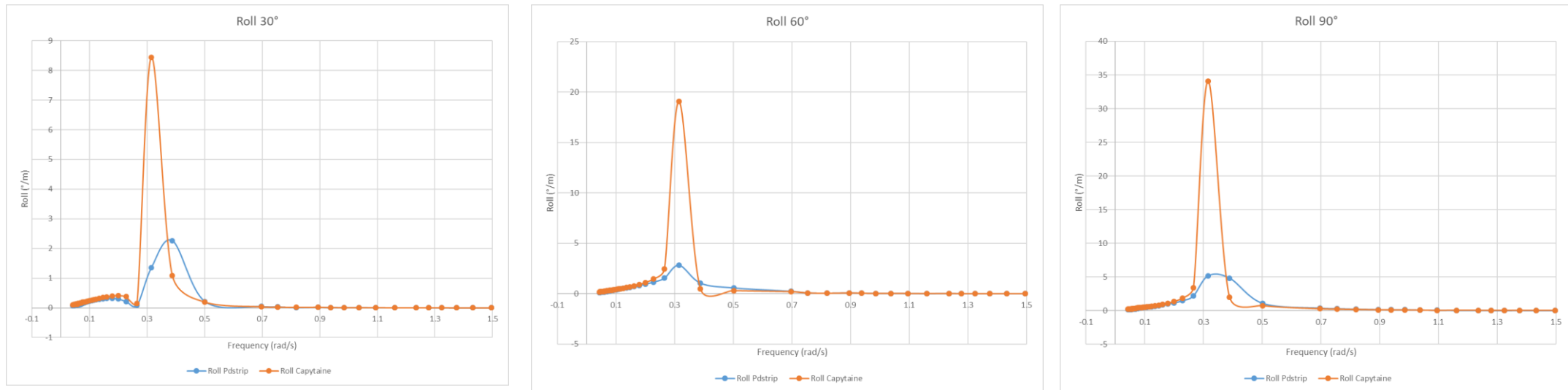


Pitch and yaw results

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The comparison with the RAOs obtained with PDSTRIP shows:

- Differences for the roll results



Roll results

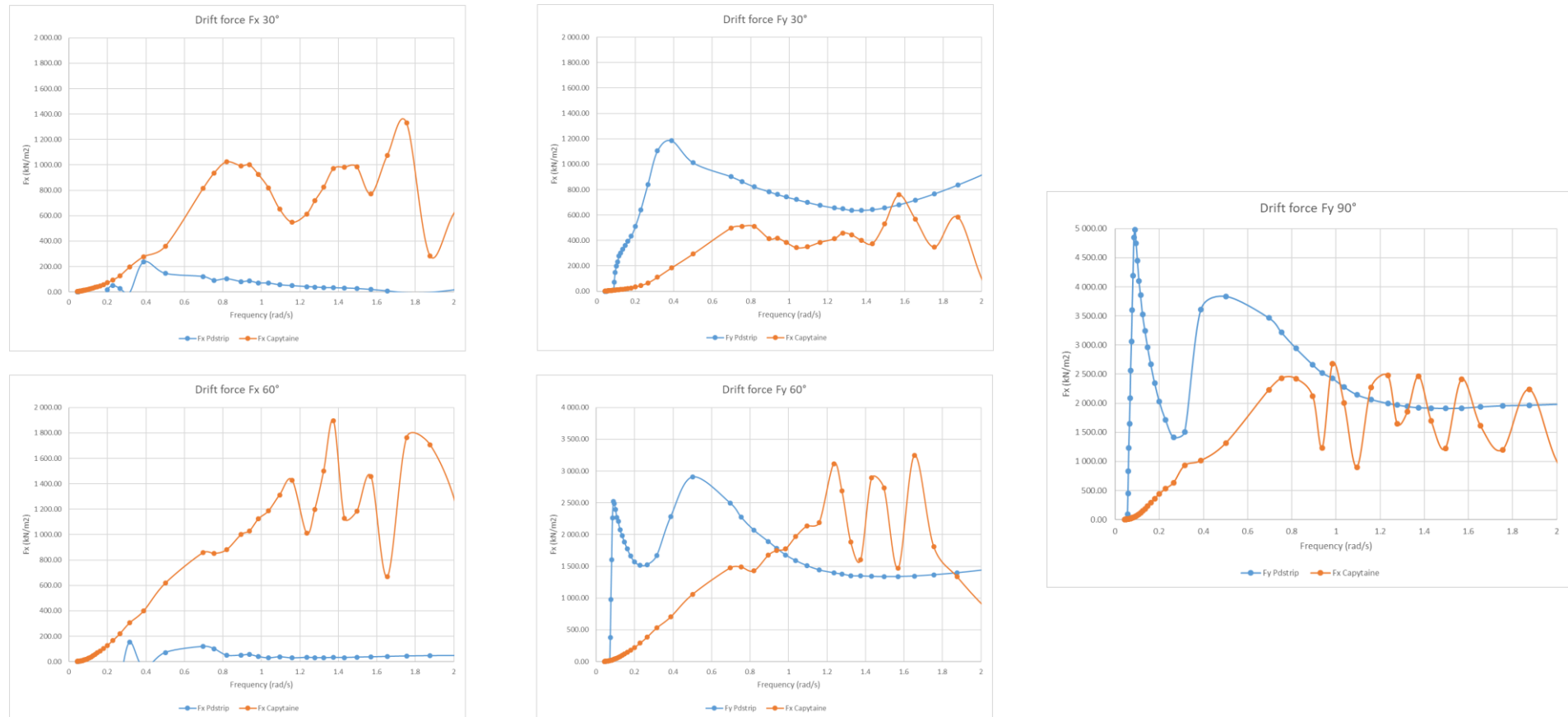
The differences between the roll results appear mainly around the resonance period, i.e. between 16 and 20 s. They would essentially result from greater radiation damping for the PDSTRIP program around this period.

No comparison has been presented for the surge because as indicated above, the PDSTRIP program does not calculate radiation damping for this dof. However, for the SIMBAD ship mooring program, an estimate of this damping is made.

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A module was created for the calculation of the drift forces according to the "Far Field Formula" (Maruo,1960 and Newmann,1967), with implementation on the basis of the PhD G. Delhommeau thesis formula.

Comparison with some results obtained with PDSTRIP is given hereafter:



As we can see in the here above figures, the comparison reveals notable differences between the two results